

Physics-Informed Neural Networks (PINNs)

Progress and Observations

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- To design a **robust PHYSICS INFORMED neural network architecture** for solving Helmholtz equations arising in the **acoustics domain**.
- Focus on addressing **spectral bias** in PINNs.
- Improve convergence for solutions containing **high-frequency components**.

Baseline Model

- Standard PINN with fully-connected neural network.
- Sinusoidal activation / Fourier feature based input embedding.
- PDE residual minimized using physics-based loss.

Observation: Baseline model struggles to represent higher-frequency modes accurately.

Improvement 1: Increasing Frequency Span

Idea:

- Increase the frequency span of the Fourier feature embedding.
- Hypothesis: Larger span $\Rightarrow$ access to higher frequencies $\Rightarrow$ better convergence.

Observation:

- No significant improvement in convergence for 1D problems.
- Learned solution tends to concentrate energy around a **dominant spectral band**.

Improvement 2: Learnable Multi-Scale Feature Map

Idea:

- Make the frequency span / scaling parameters learnable.
- Allow the network to adapt frequency content during training.

Observation:

- No desired improvement in convergence.
- In some cases, optimization becomes less stable.

Question:

- Is making frequencies learnable can be done with any other modification?
- Should there be any structural changes in the network necessary to generate higher frequencies?

This motivated me exploring mechanisms beyond additive feature representations.

Possible Reasons for Failure

- **Spectral Bias:** Neural networks prefer low-frequency solutions.
- **Additive Representations:** Linear combinations of sinusoids do not naturally generate new frequencies.
- **Optimization Dominance:** PDE loss prioritizes smoother modes early in training.

Conclusion: Frequency access $\neq$ frequency utilization.

Can Multiplication be done?

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x) \quad (1)$$

- Multiplication naturally produces **higher harmonics**.
- Current architectures rely mainly on **additive combinations**.

Insight: To generate higher frequencies, the architecture itself may need multiplicative structure.

How Can Multiplication Be Introduced?

- Gated networks: $f(x) \cdot g(x)$
- Feature-wise modulation (FiLM layers)
- Product of sinusoidal features

These mechanisms allow implicit frequency mixing inside the network.

Observation:

- Using only $\sin(\cdot)$ limits phase flexibility.
- Including $\cos(\cdot)$ allows easy phase shifts:

$$A \sin(\omega x) + B \cos(\omega x) = C \sin(\omega x + \phi)$$

This may improve representational efficiency for oscillatory solutions.

How can regularisation be achieved?

- Spectral regularization to encourage high-frequency usage.
- Curriculum learning: gradually increase PDE difficulty or frequency content.